INCH-POUND
MIL-E-22200/11A(SH)
25 April 1989
SUPERSEDING
MIL-E-22200/11(SHIPS)
10 October 1975
(See 6.7)

MILITARY SPECIFICATION

ELECTRODES, WELDING, MINERAL-COVERED, IRON-POWDER, LOW-HYDROGEN, FOR PRODUCING HY-130 WELDMENTS TO BE HEAT-TREATED

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers low-hydrogen, iron-powder type of covered electrodes, for the fabrication and repair welding of HY-130 steel to be heat-treated.

1.2 Classification. Electrodes covered by this specification shall be of the type, class, welding position, and size shown in table I, as specified (see 6.2).

TABLE I. Electrode classification.

Type	Class	Welding position	Size diameter (inch)
MIL-14018-HT	2	Horizontal fillets and flat	3/16

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-W-10430 - Welding Rods and Electrodes; Packaging of.
MIL-E-22200 - Electrodes, Welding, Covered; General Specification for.
MIL-S-24371 - Steel Plate, Alloy, Structural, High Yield Strength (HY-130).

STANDARDS

MILITARY

MIL-STD-147 - Palletized Unit Loads.
MIL-STD-271 - Requirements for Nondestructive Testing Methods.
MIL-STD-2149 - Standard Procedures for Explosion Testing Ferrous and Non-Ferrous Metallic Materials and Weldments.

(Unless otherwise indicated, copies of federal and military specifications and standards, are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government document and publication. The following other Government document and publication form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF LABOR

Code of Federal Regulations, Title 29
Part 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

0900-LP-003-9000 - Radiographic Standards for Production and Repair Welds.

(Application for copies should be addressed to the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

E 604 - Standard Test Method for Dynamic Tear Testing of Metallic Materials. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

A4.3 - Standard Methods for Determination of the Diffusible Hydrogen Content of Martensitic, Bainitic, and Ferritic Weld Metal Produced by Arc Welding.

B4.0 - Standard Methods for Mechanical Testing of Welds. (DoD adopted)

Z49.1 - Safety in Welding and Cutting.

(Application for copies should be addressed to the American Welding Society, Inc., 550 NW LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. Electrodes furnished under this specification shall be in accordance with MIL-E-22200 and as specified herein.

3.2 Composition and iron content.

3.2.1 Composition. The chemical composition of the electrode covering, except as specified in 3.2.2, is optional with the contractor. The water content of the covering shall not exceed the percent-by-weight values specified in table II. In addition, the diffusible hydrogen level shall not exceed 3.5 milliliters per one hundred grams (mL/100 g) of deposited weld metal for welds deposited with electrodes removed from sealed manufacturer's containers.

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TABLE II. Water content.

Type	Percent by weight maximum	
	Condition A[1/]	Condition B[2/]
MIL-14018-HT	0.10	0.20

1/ Condition A - When removed from contractor's sealed container.
2/ Condition B - After 9 hours exposure at 80 percent relative humidity (R.H.) and 80 degrees Fahrenheit (°F) in accordance with 4.7.

3.2.2 Total iron content. The total iron content of the covering, including any combined iron in addition to metallic iron powder, shall be not less than 15 percent.

3.3 Groove welds.

3.3.1 Mechanical properties. The mechanical properties of groove welds shall be as specified in table III.

TABLE III. Heat-treated mechanical properties.[1/]

Type	Yield strength[2/] lb/in^2	Elongation in 1.4 inches or 2 inches minimum percent	Transverse side bend	5/8 inch dynamic tear		Explosion test series
				Energy ft/lb minimum	Temperature °F	
MIL-14018-HT	135,000 to 150,000	14	[3/]	475	30	[4/]

1/ The ultimate tensile strength and percentage reduction of area shall be recorded for information only.
2/ The yield strength shall be the average of duplicate specimens measured at 0.2 percent offset by using a class B extensometer or better. The minimum single value shall be 135,000 pounds per square inch (lb/in^2).
3/ The convex surface of the specimen after bending shall have no visual cracks exceeding 1/8 inch. The corners of the specimen shall have no visual cracks exceeding 3/16 inch. The bend radius shall be 3 times the specimen thickness (3t) (see 4.5.1).
4/ The criteria of performance shall be in accordance with the following when tested in accordance with 4.5.2:

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(a) Two crack-starter weldments:

(1) First shot - Crack-starter bead may crack. Percent reduction in thickness obtained for information only. No piece shall be thrown out of weldments. No through-thickness cracks shall be present. No crack shall extend into hold-down area (that area outside a 12-1/2 inch radius measured from the center of the weldment).

(2) Second shot - Percent reduction in thickness obtained for information only. No piece shall be thrown out of weldment. Through-thickness cracks are acceptable. No cracks shall extend into hold-down area (that area outside a 12-1/2 inch radius measured from the center of the weldment).

3.4 Chemical composition of deposited weld metal. Chemical composition of deposited weld metal shall be as specified in table IV.

TABLE IV. Chemical composition (weight-percent)[1/] of deposited weld metal.

Type	Element	Weight-percent
MIL-14018-HT	Carbon	0.10
	Manganese	.40 to .80
	Phosphorus	.013
	Sulfur	.010
	Silicon	.50
	Nickel	7.00 to 9.00
	Chromium	.05 to .60
	Molybdenum	.40 to .70
	Vanadium	.04 to .10

1/ Single values are maximum percentages.

3.5 Electrode identification. Electrodes shall be identified as specified in MIL-E-22200.

3.5.1 Heat and lot identification. When specified (see 6.2), each electrode shall be marked with heat lot or other controlled marking code identification as specified in MIL-E-22200.

3.6 Electrode length. The electrode length shall be 14 ± 1/4 inches.

3.7 Core wire alloy identity. The core wire alloy identity requirements shall be as specified in MIL-E-22200.

3.8 Covered electrode alloy identity. When specified (see 6.2), the covered electrode alloy identity requirements shall be as specified in MIL-E-22200.

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3.9 Soundness. Electrodes shall deposit weld metal which conforms to the radiographic acceptance standards in accordance with class I of NAVSEA 0900-LP-003-9000.

4. QUALITY ASSURANCE PROVISIONS

4.1 The quality assurance provisions shall be in accordance with MIL-E-22200 and as specified herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification tests shall be in accordance with MIL-E-22200, table V and 4.5.3 herein. Schedule A tests of table V shall be conducted by the manufacturer. Upon successful completion of schedule A tests, schedule B welding and testing will be conducted by the Government.

TABLE V. Summary of weld metal tests required for qualification. 1/, 2/

Electrode type	Test schedule	Examination	Test procedures	Requirements
-	A	Radiographic	4.5.1, figures 1 and 2	3.9
		Tensile	4.5.1, figures 1 and 2	table III
		Bend	4.5.1 and figure 1	table III
		Chemical	MIL-E-22200	table IV
		Dynamic tear	4.5.1, figures 1 and 2	table III
MIL-14018-HT	B	Explosion test series	4.5.2 and MIL-STD-2149	table III
		Radiographic	4.5.2.2	3.9
		Tensile	4.5.2.2	table III
		Bend	4.5.2.2	table III
		Dynamic tear	4.5.2.2 and ASTM E 604	table III

1/ The electrode sample selected in accordance with MIL-E-22200 shall be large enough to make all welds required in schedule A and B tests.
2/ For adjunct testing requirements, see table VII.

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4.4 Quality conformance inspection.

4.4.1 Lot. A lot of electrodes shall be as specified in MIL-E-22200 for inspection level A.

4.4.2 Sampling for quality conformance inspection. Sampling for quality conformance inspection shall be in accordance with MIL-E-22200, except that electrodes for the covering moisture or the diffusible hydrogen test taken from the production line shall be packaged for delivery to the test site. Any packages taken from stock shall be unopened when delivered to the test site.

4.4.3 Quality conformance inspection. The samples selected in accordance with 4.4.2 shall be subjected to the weld metal tests specified in table VI, the applicable examination and tests specified in table VII, and the packaging inspection specified in 4.9 (see 6.3 and figure 3).

TABLE VI. Summary of weld metal tests required for quality conformance inspection.1/

Electrode type	Examination	Test procedures	Requirements
MIL-14018-HT	Radiographic	4.5.1 and figure 1	3.9
	Tensile	4.5.1 and figure 1	table III
	Chemical	MIL-E-22200	table IV
	Dynamic tear	4.5.1 and figure 1	table III

1/ For adjunct testing requirements, see table VII.

4.4.3.1 Unsatisfactory test results. If the results of the first tests, except dynamic tear, are determined to be unsatisfactory, two retests will be permitted. The results of both retests shall be satisfactory for lot acceptance. Retesting of the 5/8-inch dynamic tear test specimens shall be in accordance with ASTM E 604.

4.5 Weld metal test procedures.

4.5.1 One-inch and 4-inch thick weld test. For the dimensions of the test weldments and the location of the tensile and dynamic tear specimens, see figures 1 and 2. The 1-inch thick test weldment (see figure 1) shall be extended to provide for two transverse side bend test specimens when qualification testing an electrode. The transverse side bend test specimens shall be taken next to the discard ends. The test assemblies shall be welded, heat treated, radiographed, and tested as follows:

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(a) The base-plate material shall be HY-130 steel in accordance with MIL-S-24371.
(b) Direct current (dc) reverse polarity shall be used.
(c) Welding shall be done in the flat position.
(d) The joint surfaces shall not be clad or buttered.
(e) The preheat and interpass temperature range shall be 250 to 275°F.
(f) The heat input shall be 38,000 to 42,000 joules per inch.
(g) Welding may be continuous except for interpass cooling. Planned interruption of welding shall be followed immediately by one of the following heat soaking treatments without the weldment cooling below the interpass temperature:

(1) Partially completed welds 1/2-inch thick or less at 250 plus 25 minus 0°F for 4 hours minimum.
(2) Partially completed welds over 1/2-inch thick at 350 plus 25 minus 0°F for 12 hours minimum.

(h) The weldment shall be given the following heat treatment without cooling below the interpass temperature range:

(1) Charge into furnace at 250 to 500°F.
(2) Heat to 1700 ± 25°F, hold at temperature for 1 hour per inch of thickness and water quench.
(3) Reheat to 1500 ± 25°F, hold at temperature for 1 hour per inch of thickness and water quench.
(4) Reheat to 1100 ± 25°F, hold at temperature for 1 hour per inch of thickness and water quench.

When it is planned to cool the weldment to room temperature before heat-treating, the weldment shall be given a post weld soaking treatment of 350 plus 25 minus 0°F for 12 hours minimum without the weldment cooling below the minimum interpass temperature.

(i) The heat-treated weldments shall then have the reinforcement and backing strip removed flush with the base plate on both surfaces.
(j) The flush welds shall be inspected radiographically in accordance with level 2-2T of MIL-STD-271 to determine compliance with 3.9.
(k) The bend test specimens, when required, and the tension test specimens shall be machined and tested in accordance with AWS B4.0 to determine compliance with table III.
(l) The 5/8-inch dynamic tear test specimens shall be machined and tested in accordance with ASTM E 604 to determine compliance with table III.

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4.5.2 Explosion test series (crack starter) and mechanical properties prolongation tests.

4.5.2.1 Welding parameters. Fabrication of the explosion crack starter and mechanical property prolongation assemblies shall be as follows:

(a) The base-plate material shall be HY-130 steel in accordance with MIL-S-24371.
(b) Welding shall be done in the flat position using (dc) reverse polarity.
(c) The joint surfaces shall not be clad or buttered.
(d) Upon removal from manufacturer's unit containers or holding oven, electrodes shall be placed in a holding oven at the job site. The temperature of the holding oven shall be 225 to 300°F. The welder shall use each electrode immediately upon removal from the holding oven at the job site.
(e) The preheat and interpass temperature range shall be 250 to 275°F.
(f) The heat input shall be 38,000 to 42,000 joules per inch.
(g) Welding may be continuous except for interpass cooling. Planned interruption of welding shall be followed immediately by one of the following heat soaking treatments without tube weldment cooling below the interpass temperature:

(1) Partially completed welds 1/2-inch thick or less at 250 plus 25 minus 0°F for 4 hours minimum.
(2) Partially completed welds over 1/2-inch thick at 350 plus 25 minus 0°F for 12 hours minimum.

(h) The weldment shall be given the following heat treatment without cooling below the interpass temperature range:

(1) Charge into furnace at 250 to 500°F.
(2) Heat to 1700 ± 25°F, hold at temperature for 2 hours and water quench.
(3) Reheat to 1500 ± 25°F, hold at temperature for 2 hours and water quench.
(4) Reheat to 1100 ± 25°F, hold at temperature for 2 hours and water quench.

When it is planned to cool the weldment to room temperature before heat treating, the weldment shall be given a post weld soaking treatment of 350 plus 25 minus 0°F for 12 hours minimum without the weldment cooling below the minimum interpass temperature.

(i) The heat-treated weldment shall have the reinforcement left in place.
(j) With the weld reinforcement left in place, 100 percent of the welds shall be inspected radiographically in accordance with level 2-2T of MIL-STD-271 to determine compliance with 3.9 herein.

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(k) The explosion test series (crack starter) weldments shall be tested only after the satisfactory completion of the following tests:

(1) Small scale mechanical tests from the two prolongations shall be in accordance with MIL-STD-2149.
(2) The radiographic tests as specified in (j) above.

(l) The explosion test series (crack starter) weldments, when tested, shall have the weld reinforcement left in place.

4.5.2.2 The two crack starter test plates and mechanical prolongations shall be fabricated and tested in accordance with schedule B of table V.

4.5.3 Other tests. Table VII summarizes tests, in addition to those for qualification testing of electrodes and for quality conformance inspection, as applicable.

TABLE VII. Summary of other tests.

Tests	Test procedures	Requirements
Visual and dimensional	MIL-E-22200	MIL-E-22200 and 3.6
Concentricity	MIL-E-22200	MIL-E-22200
Dielectric strength[1/]	MIL-E-22200	MIL-E-22200
Covering, flaking and cracking[1/]	MIL-E-22200	MIL-E-22200
Covering moisture	MIL-E-22200 and 4.7	3.2.1
Total iron content in covering[1/]	MIL-E-22200	3.2.2
Alloy identity - core wire	MIL-E-22200	MIL-E-22200
Alloy identity - finished electrode[2/]	MIL-E-22200 and 4.8	3.8
Diffusible hydrogen[3/]	4.6	3.2.1

1/ Required for qualification only.
2/ Required when specified (see 6.2).
3/ Required for quality conformance testing.

4.6 Diffusible hydrogen. A sample of every lot of electrodes furnished to the requirements of this specification shall be tested for diffusible hydrogen level in the weld metal (see 6.3 and figure 3). The hydrogen level of the deposited weld metal shall be determined by the analytical method of gas chromatography or by the method of collection over mercury as specified in AWS A4.3.

4.7 Temperature and humidity environmental exposure procedures. The electrode sample shall be exposed in a blue M environmental chamber model FR 251-1 or equal for 9 hours minimum at 80 minus 0 plus 5°F and R.H. 80 minus 0 plus 5 percent. The chamber shall have a two-pen recorder with the first pen for dry

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bulb temperature and the second pen for the differential between wet bulb and dry bulb temperatures. The chamber is available in all standard voltages for the various power supplies. The procedure shall be as follows:

(a) The electrode sample in unopened packages shall be heated to a temperature minus 0°F, plus 10°F above the dew point of the chamber at the time of loading.
(b) The electrode sample shall be loaded into the chamber without delay after the packages are opened.
(c) The electrodes shall be placed in the chamber in a vertical position on 1-inch centers.
(d) Time, temperature, and humidity shall be continuously recorded for the period that the electrodes are in the chamber.
(e) Counting of exposure time shall start when the required temperature and humidity in the chamber are established.
(f) At the end of the exposure time, the electrodes shall be removed from the chamber and a sample of the coating taken for moisture determination.

4.8 Alloy identity certification. When specified (see 6.2), the Government representative will affix his stamp and date on the container, containing electrodes in accordance with MIL-E-22200. The letters "AT", to indicate alloy tested, shall be affixed to the container adjacent to the date.

4.8.1 The identification of melts, heats or lots; the general requirements for test specimens; the rejection of nonconforming lots or specimens; and the retest and resubmittal of nonconforming lots shall be in accordance with MIL-E-22200.

4.9 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Preservation. Preservation of electrodes shall be level A.

5.1.1 Level A. Electrodes shall be preserved in class 1 unit containers in accordance with MIL-W-10430.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Levels A and B. Electrodes shall be packed level A or B in accordance with MIL-W-10430, except that for levels A and B packing, palletization for other than cylindrical cans shall be in accordance with MIL-STD-147.

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5.2.2 Level C. Packing shall be accomplished in a manner which will ensure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity. The shipping containers or method of packing shall be in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Interior packages. Shipment marking information shall be provided on interior packages in accordance with the manufacturer's commercial practice. In addition to the lot identification as specified in MIL-E-22200, the information shall include MIL type, size, specification number, manufacturer's or distributor's name, date of manufacture (month and year), and manufacturer's or distributor's brand or type designation. Packages, or the smallest integral unit within a shipping container, shall carry the following warning label, as a minimum, prominently displayed in legible type on the package:

WARNING: Protect yourself and others. Read and understand this label.

FUMES AND GASES can be dangerous to your health. ARC RAYS can injure eyes and burn skin. ELECTRIC SHOCK can kill.

Read and understand the manufacturer's instructions and your employer's safety practices.
Keep your head out of the fumes.
Use enough general ventilation or exhaust or both to keep fumes and gases from your breathing zone, and the general area.
Wear correct eye, ear and body protection.
Do not touch live electrical parts.
See American National Standard Z49.1 "Safety in Welding and Cutting" published by the American Welding Society Inc., 550 NW LeJeune Road, P.O. Box 351040, Miami, FL 33135; OSHA Safety and Health Standards, 29 CFR 1910 are for sale by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

DO NOT REMOVE THIS LABEL

5.3.2 Exterior shipping containers and palletized unit loads. Shipment marking information shall be provided on exterior shipping containers and palletized unit loads in accordance with manufacturer's commercial practice. In addition to the information required for interior packages as specified in 5.3.1 (except the caution label), the information shall include shipping destination, stock number as listed by contract, and contracting activity's order number, item number, and name.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification is intended to cover nickel-chromium-molybdenum-vanadium alloy steel mineral-covered welding electrodes depositing weld metal, that in the heat-treated condition meets the mechanical properties specified herein when welded on HY-130 steel with the shielded metal-arc process using direct current reverse polarity and the techniques specified herein. The primary use of these electrodes is for the welding of HY-130 steel for applications in the heat-treated condition.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) Type, class, and size required (see 1.2).
(d) When heat or lot identification is required for individual electrodes (see 3.5.1).
(e) Whether alloy identity testing of the core wire of each covered electrode after final marking is required (see 3.8, 4.8, and footnote 2 to table VII).
(f) Level of packing required (see 5.2).
(g) Electrodes should be ordered by the pound.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.4.3 and 4.6	DI-MISC-80678	Certification data/ report	----

The above DIDs were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

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6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List No. 22200 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests must be made in accordance with "Provisions Governing Qualification SD-6" (see 6.4.1).

6.4.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.5 Inspection after delivery. Post delivery inspection of electrodes to determine conformity to this specification and for acceptance thereof is the responsibility of the contracting activity.

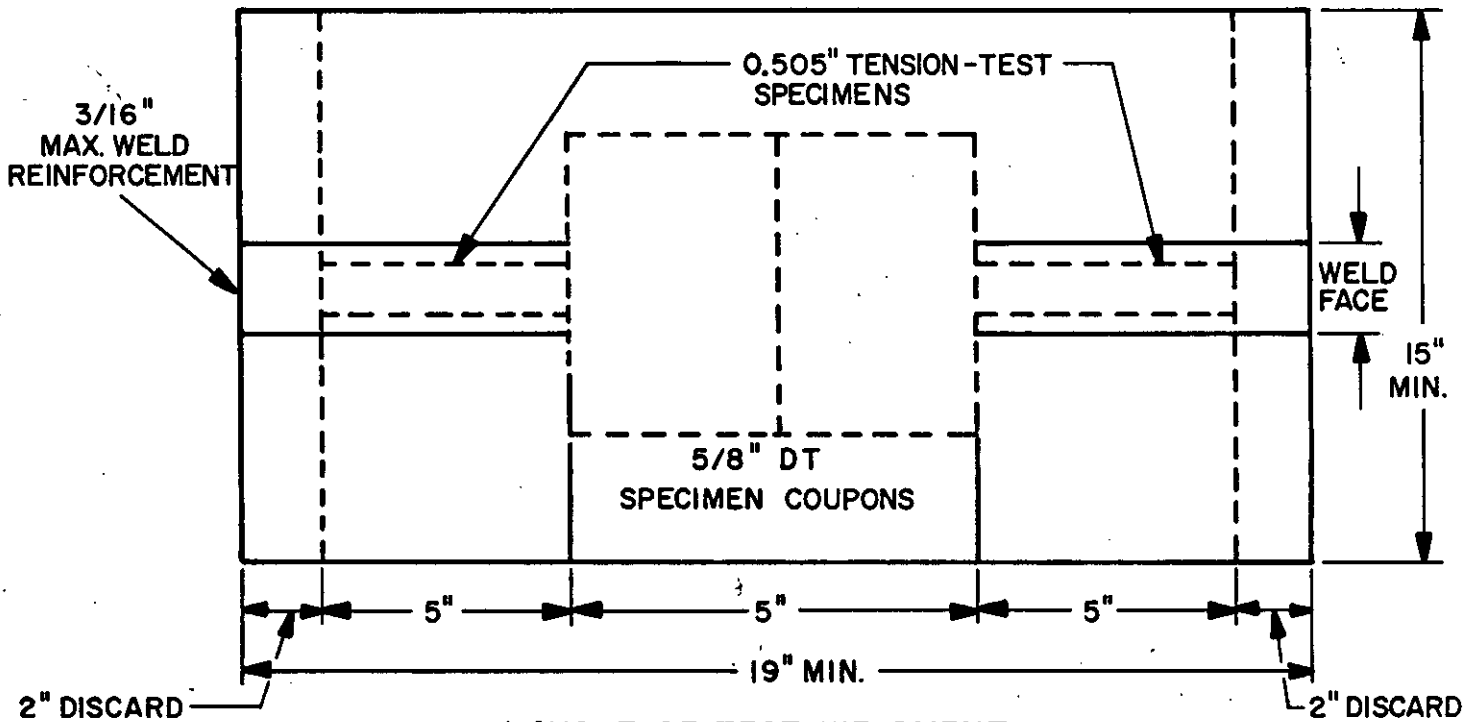
6.6 Subject term (key word) listing.

Covering moisture
Crack-starter weldment
Diffusible hydrogen
Hold-down area

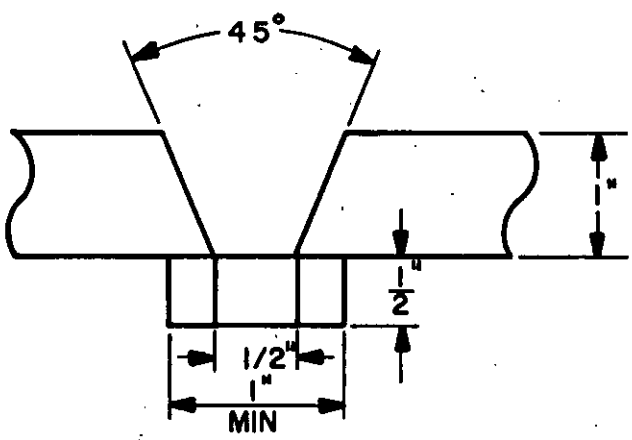
6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 3439-N601)

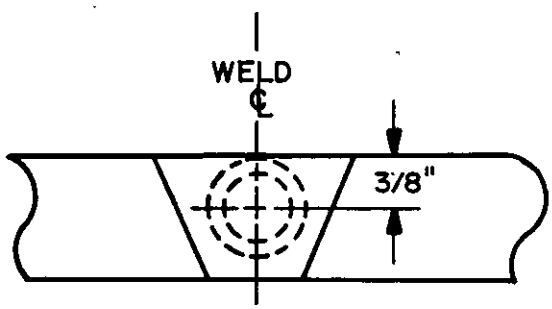
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LAYOUT OF TEST WELDMENT



JOINT GEOMETRY



SPECIMEN LOCATION

SH 10868

FIGURE 1. One-inch thick weld test.

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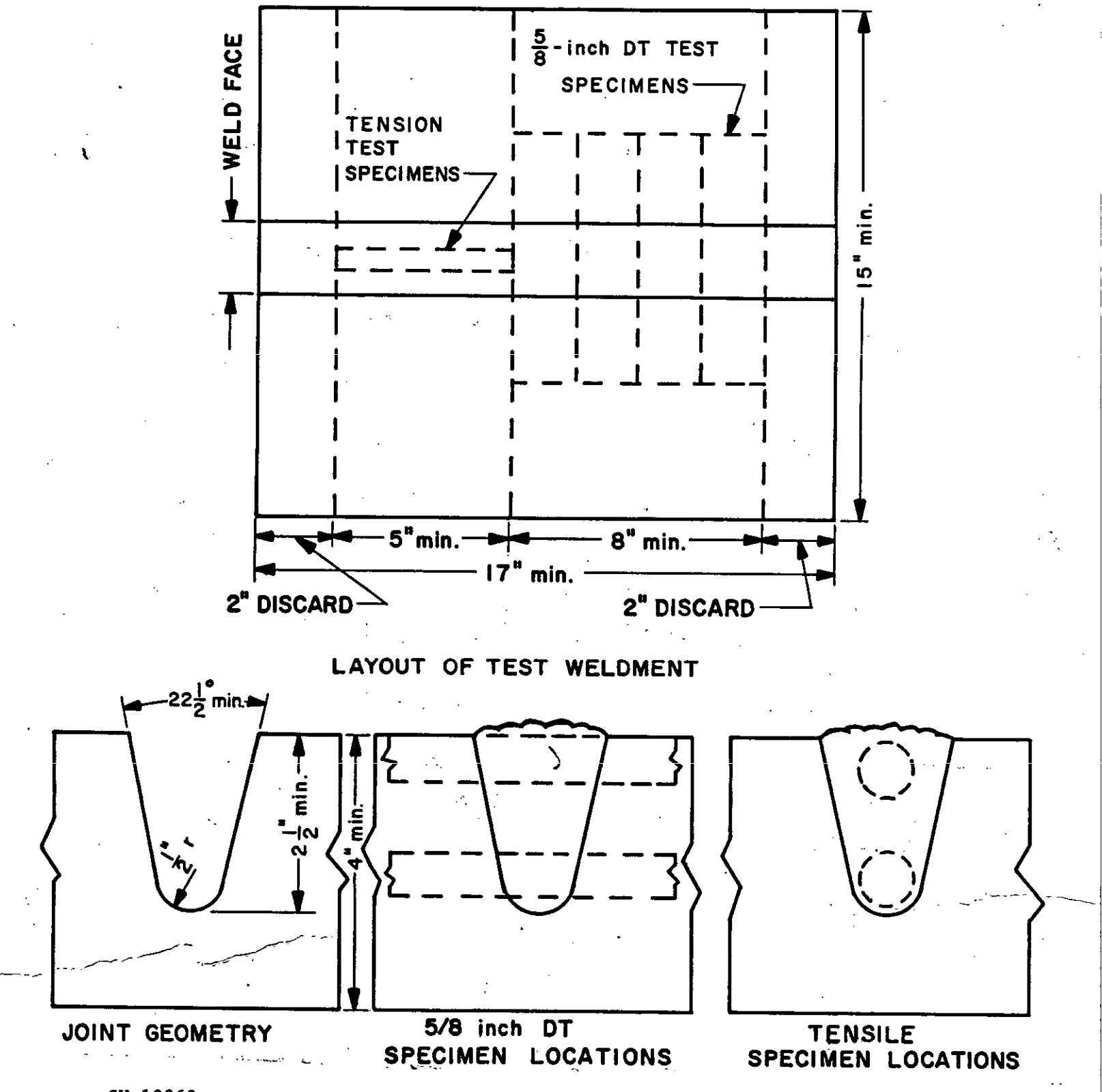


FIGURE 2. Four-inch thick weld test.

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CERTIFICATION OF QUALITY CONFORMANCE TESTS

Manufacturer or distributor _____

Address _____

Date _____

Specification MIL- _____

Type MIL- _____

Diameter and length _____

Lot number _____

Chemical analysis (Complete)

Carbon _____

Manganese _____

Silicon _____

Phosphorus _____

Sulfur _____

Chromium _____

Nickel _____

Molybdenum _____

Vanadium _____

_____

Chemistry was taken from: Chem. pad
Groove weld

Concentricity (percent) _____

Covering moisture
Condition A _____
Condition B _____

Customer's name _____

Customer's order number _____

Core wire heat number _____

Lot identification MIL-E-22200 _____
paragraph _____

Wet mix number _____

Mechanical test results

Yield strength _____
(0.2 percent offset method)

Tensile strength _____

Elongation _____

Reduction of area (percent) _____

Impact test results

Dynamic tear
Impact energy

1 _____
2 _____

Grinding during test plate preparation
(see MIL-E-22200) _____

Operator error
(layer numbers) _____

SAMPLE

FIGURE 3. Sample certification of quality conformance.

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Radiographic _____

Diffusible hydrogen mL/g of deposited
weld metal test specimen number
Average value _____
Minimum single value _____
Test/Analysis Method: Gas chromatography , over mercury

We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements.

_____
Authorized contractor representative

SAMPLE

FIGURE 3. Sample certification of quality conformance - Continued.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-22200/11A(SH)	ELECTRODES, WELDING, MINERAL-COVERED, IRON-POWDER, LOW-HYDROGEN, FOR PRODUCING HY-130 WELDMENTS TO BE HEAT-TREATED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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